

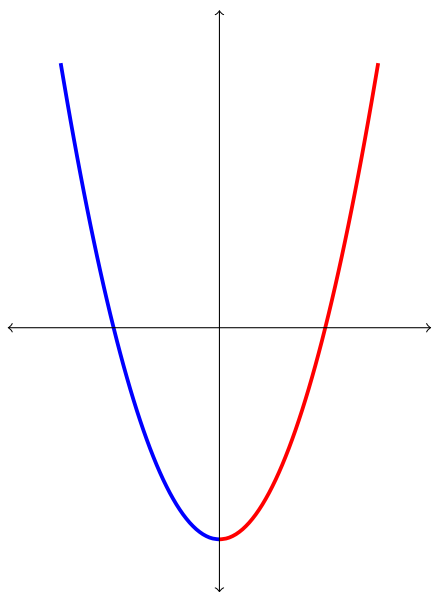
WEEK 3: LECTURE NOTE

PROPERTIES OF GRAPHS

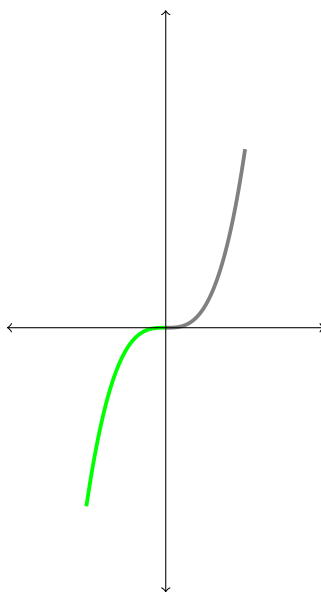
Definition 1. (*Symmetry of Graphs*)

1. A graph has **symmetry with respect to the y -axis** if whenever (x, y) is on the graph, then so is the point $(-x, y)$. A function whose graph is symmetric with respect to the y -axis is called an *even function*.
2. A graph has **symmetry with respect to the origin** if whenever (x, y) is on the graph, then so is the point $(-x, -y)$. A function whose graph is symmetric with respect to the origin is an *odd function*.
3. A graph has **symmetry with respect to the x -axis** if whenever (x, y) is on the graph, then so is the point $(x, -y)$. A graph that is symmetric with respect to the x -axis is **not** the graph of a function except for the graph of $y = 0$.

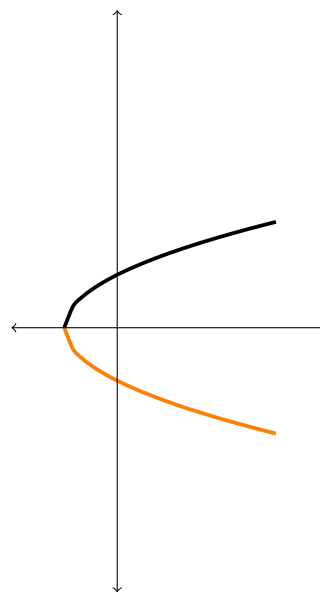
These three types of symmetry are illustrated in the Figures below.



Symmetric with
respect to y -axis



Symmetric with
respect to Origin



Symmetric with
respect to x -axis

Definition 2. (*EVEN AND ODD FUNCTIONS*)

1. A function f is said to be **even** when, for each x in the domain of f , $f(-x) = f(x)$.
2. A function f is said to be **odd** when, for each x in the domain of f , $f(-x) = -f(x)$.

Example 3 (EVEN AND ODD FUNCTIONS). Determine whether each function is even, odd, or neither.

(a) $g(x) = x^3 - x$

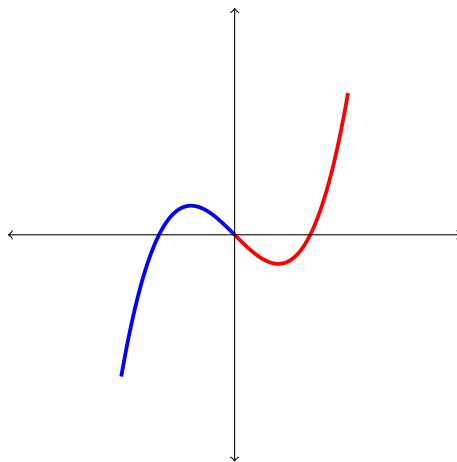
(b) $h(x) = x^2 + 1$

(c) $f(x) = x^3 - 1$

Solution. We will show here algebraically and graphically.

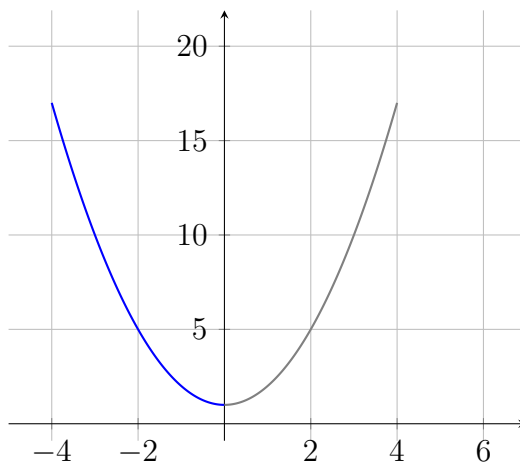
(a) The function g is odd because

$$g(-x) = (-x)^3 - (-x) = -x^3 + x = -(x^3 - x) = -g(x).$$



The graph is symmetric with respect to the Origin, and therefore g is an odd function.

(b) The function h is even because $h(-x) = (-x)^2 + 1 = x^2 + 1 = h(x)$.

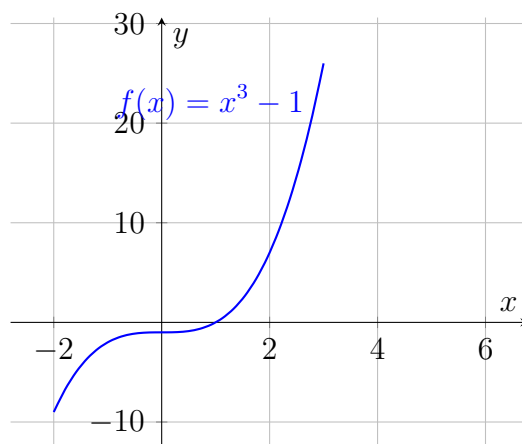


The graph is symmetric with respect to the y -axis, and therefore h is an even function.

(c) Substituting $-x$ for x produces

$$f(-x) = (-x)^3 - 1 = -x^3 - 1.$$

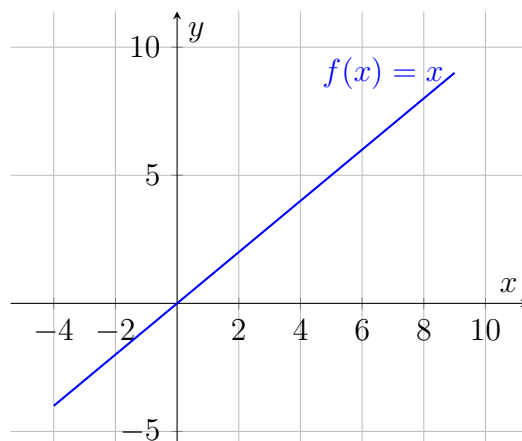
Because $f(x) = x^3 - 1$ and $-f(x) = -x^3 + 1$ we can conclude that $f(-x) \neq f(x)$ and $f(-x) \neq -f(x)$. So, the function f is neither even nor odd.



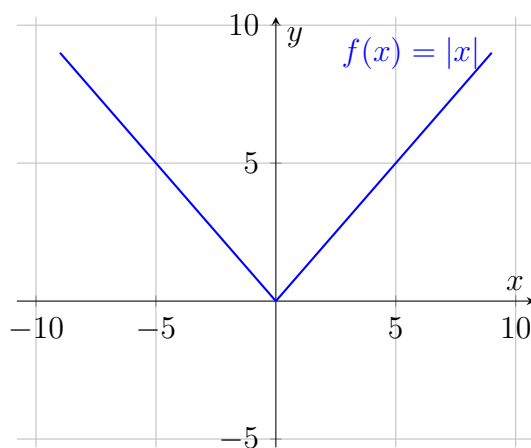
The graph is neither symmetric with respect to y -axis nor symmetric with respect to origin; therefore f is neither even nor odd function.

Summary of Graphs of Parent Functions

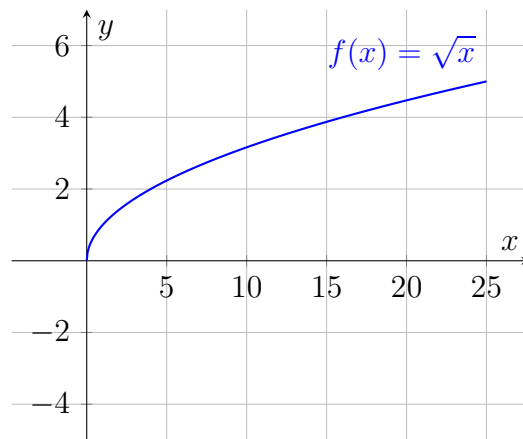
1. Linear Function $f(x) = x$



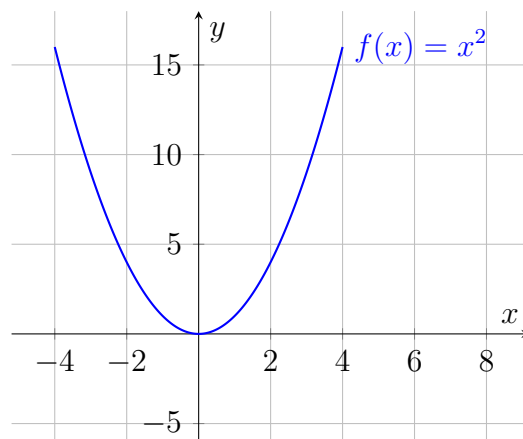
2. Absolute Value Function $f(x) = |x|$



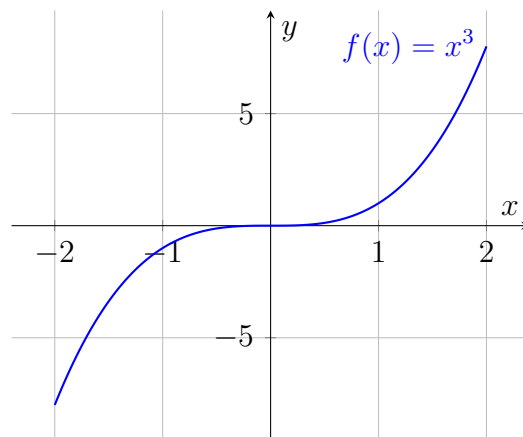
3. **Square Root Function** $f(x) = \sqrt{x}$



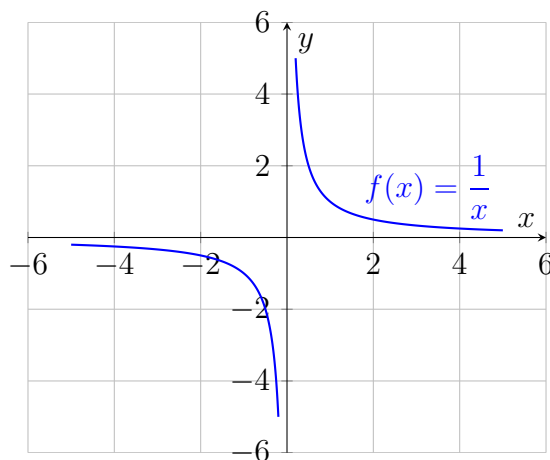
4. **Quadratic Function** $f(x) = x^2$



5. **Cubic Function** $f(x) = x^3$



6. Rational Function $f(x) = \frac{1}{x}$



Shifting, reflecting, and stretching graphs

Shifts, stretches, shrinks, and reflections are called **transformations**. Many graphs of functions can be created from combinations of these transformations.

1. Vertical and Horizontal Shifts

Let c be a positive real number. Vertical and horizontal shifts in the graph of $y = f(x)$ are represented as follows.

- (a) Vertical shift c units upward: $h(x) = f(x) + c$
- (b) Vertical shift c units downward: $h(x) = f(x) - c$
- (c) Horizontal shift c units to the right: $h(x) = f(x - c)$
- (d) Horizontal shift c units to the left: $h(x) = f(x + c)$

2. Reflections in the Coordinate Axes

Reflections in the coordinate axes of the graph of $y = f(x)$ are represented as follows.

- (a) Reflection in the x -axis: $h(x) = -f(x)$
- (b) Reflection in the y -axis: $h(x) = f(-x)$

Rigid Transformations: Shifting and Reflections

Example 4 (Shifts in the Graph of a Function). *Compare the graph of each function with the graph of $f(x) = x^3$.*

- (a) $g(x) = x^3 - 1$
- (b) $h(x) = (x - 1)^3$
- (c) $k(x) = (x + 2)^3 + 1$

Solution.

- (a) Vertical Shift one unit downward

We obtain the graph of g by shifting the graph of f one unit downward.

- (b) Horizontal shift one unit to the right

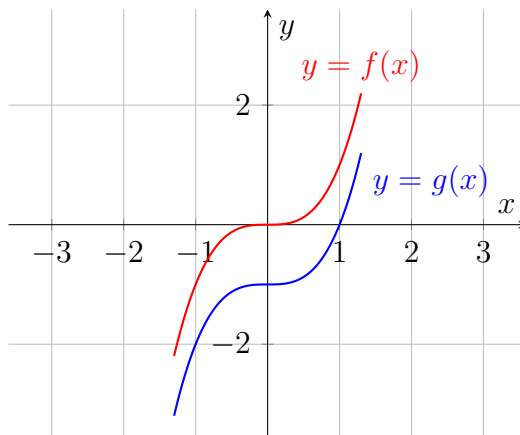
We obtain the graph of h by shifting the graph of f one unit to the right.

- (c) Horizontal and Vertical shifts

We obtain the graph of k by shifting the graph of f two units to the left and then one unit upward.

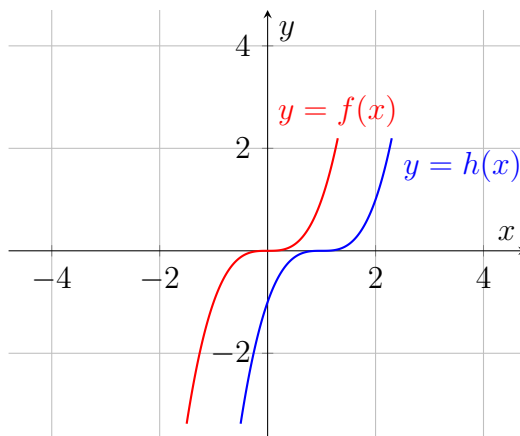
Example 5. (*Rigid Transformations*)

- (a) Graphs of $f(x) = x^3$ and $g(x) = x^3 - 1$



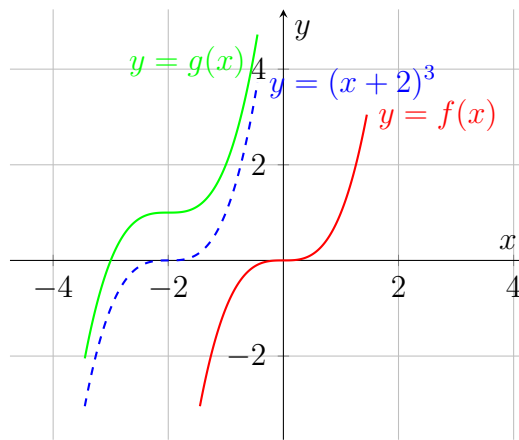
Vertical shift: one unit downward

- (b) Graphs of $f(x) = x^3$ and $h(x) = (x - 1)^3$



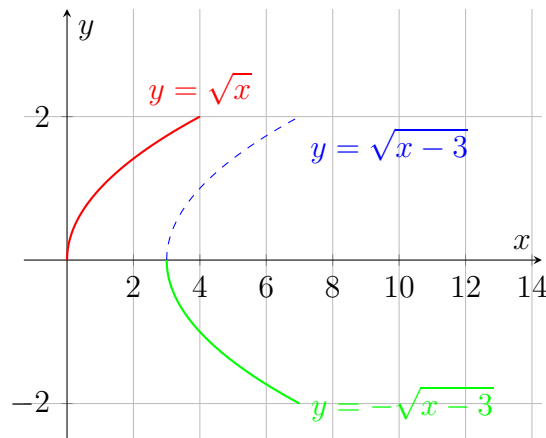
Horizontal shift: one unit right

- (c) Graphs of $f(x) = x^3$ and $k(x) = (x + 2)^3 + 1$



Two units left and one unit upward

Example 6 (Reflections and Shifts). Compare the graph of $g(x) = -\sqrt{x-3}$ function with the graph of $f(x) = \sqrt{x}$.



Shift three units right and then reflect in the x-axis;

OR

Reflect in the x-axis and then shift three units right.

Example 7 (Reflections and Shifts). Sketch the graph of $h(x) = 3 - (x + 2)^2$.

Solution. The parent function is $f(x) = x^2$. Then

$$h(x) = 3 - (x + 2)^2 = 3 - f(x + 2).$$

We apply the following sequence of transformations on the graph of the parent function f to obtain the graph of h .

(i) The parent function $f(x)$ is **Reflected about the x-axis**; that is,

$$k(x) = -f(x).$$

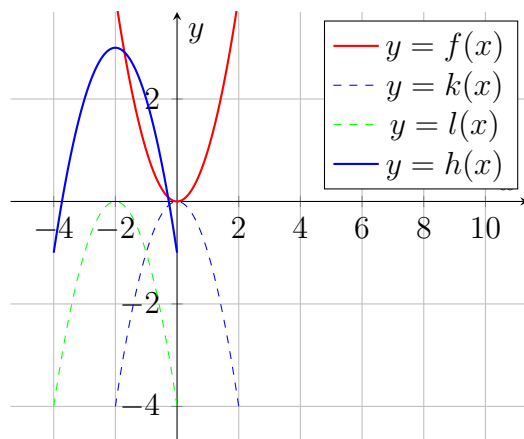
(ii) $k(x)$ is shifted horizontally two units to the left; that is,

$$l(x) = k(x + 2) = -f(x + 2).$$

(iii) Finally $l(x)$ is shifted vertically three units upward to obtain h ; that is,

$$h(x) = 3 + l(x) = 3 - k(x) = 3 - f(x + 2) = 3 - (x + 2)^2.$$

The graph of h with the graphs of f , k and l is given below.



Reflect in the x -axis $\rightarrow$ Shift 2 units left $\rightarrow$ Shift 3 units upward

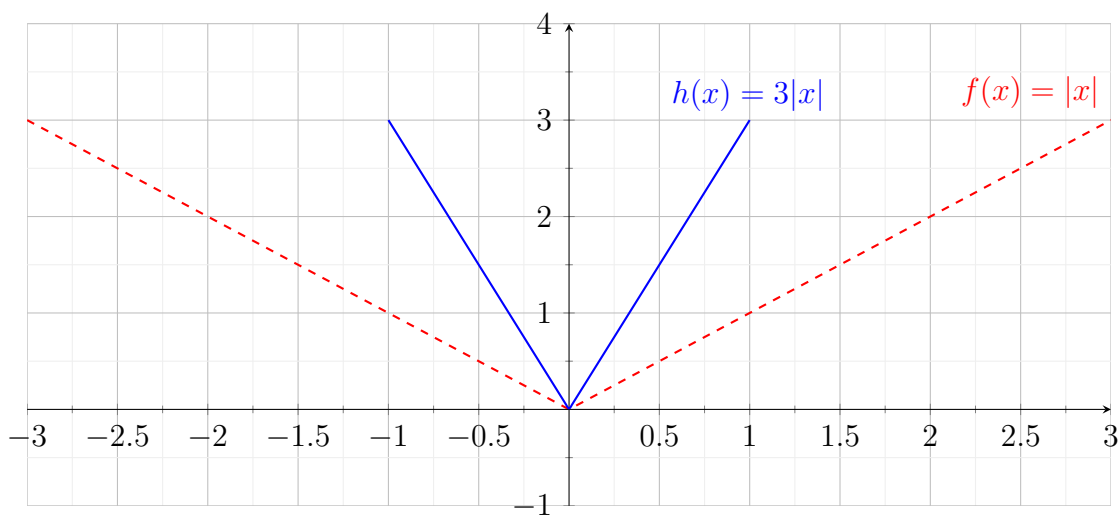
- **Rigid transformation:** We have seen that horizontal shifts, vertical shifts, and reflections change only the *position* of the graph in the coordinate plane with the basic shape of the graph unchanged. These transformations are called rigid transformations.
- **Nonrigid transformations:** Transformations that cause a *distortion* - a change in the shape of the original graph- are called nonrigid transformations.

A nonrigid transformations of the graph of $y = f(x)$

1. **VERTICAL STRETCH:** $g(x) = cf(x)$ for $c > 1$
2. **VERTICAL SHRINK:** $g(x) = cf(x)$ for $0 < c < 1$.
3. **HORIZONTAL STRETCH:** $h(x) = f(cx)$ when $0 < c < 1$.
4. **HORIZONTAL SHRINK:** $h(x) = f(cx)$ for $c > 1$.

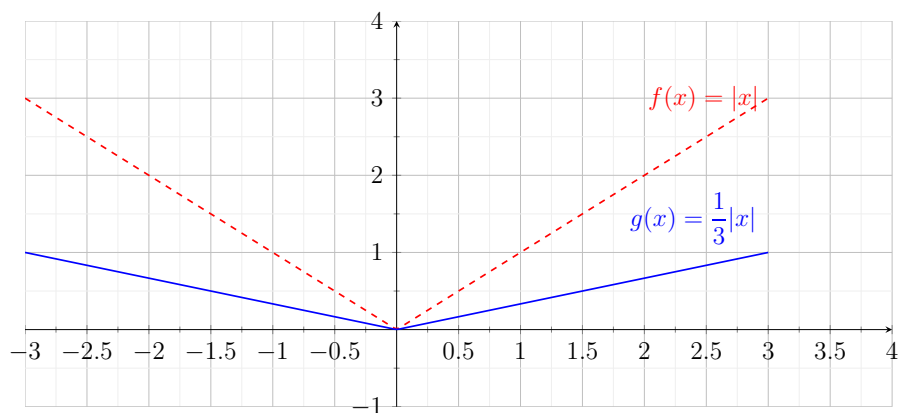
Example 8 (Nonrigid Transformations). Compare the graph of the function $h(x) = 3|x|$ with the graph of $f(x) = |x|$.

Solution. The graph of $h(x) = 3|x| = 3f(x)$ is a vertical stretch (each y -value is multiplied by 3) of the graph of f .

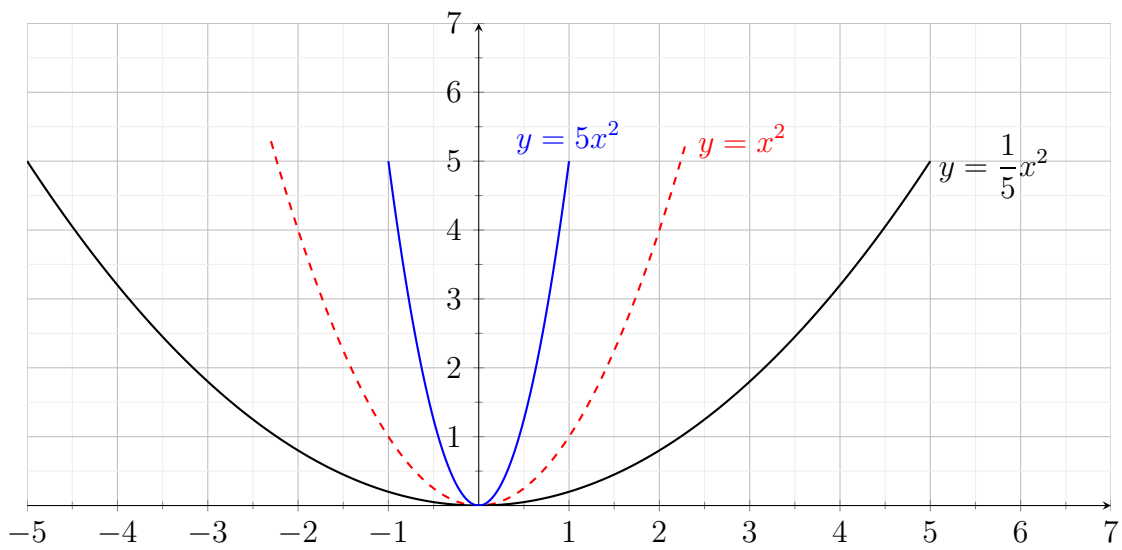


Example 9 (Nonrigid Transformations). Compare the graph of the function $g(x) = \frac{2}{3}|x|$ with the graph of $f(x) = |x|$.

Solution: The graph of $g(x) = \frac{1}{3}|x| = \frac{1}{3}f(x)$ is a vertical shrink (each y -value is multiplied by $\frac{1}{3}$) of the graph of f .



Example 10 (Nonrigid Transformations). Compare the graphs of the functions $g(x) = \frac{1}{5}x^2$ and $h(x) = 5x^2$ with the graph of $f(x) = x^2$.



Example 11 (Nonrigid Transformations). Compare the graph of the function $f(x) = 4x^2 - 8x$ with the graph of $g(x) = f(\frac{1}{2}x)$.

Note that $g(x) = f(\frac{1}{2}x) = 4(\frac{1}{2}x)^2 - 8(\frac{1}{2}x) = x^2 - 4x = (x - 2)^2 - 4$.

